Optimization for Deep Learning (OPT4DL)

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Overparameterization and Interpolation

Overparameterization of DL models



- Modern DL models are overparameterized, that is, if $\mathrm{w} \in \mathbb{R}^d$ denotes the model parameters and n denotes the number of training data points, then $d > n$
- Thus, we have more unknowns than knowns and consequently multiple global minima for our training task

$$\min_w f(w) = \frac{1}{n} \sum_{i=1}^{n} \ell(w, z_i) \tag{1}$$

- Effectively, there are parameters w^* that can interpolate all samples

$$w^* = \arg\min_w f(w) = \frac{1}{n} \sum_{i=1}^{n} \ell(w, z_i) \qquad \Leftrightarrow \qquad w^* = \arg\min_w \ell(w, z_i), \quad \forall i \in \{1, \ldots, n\} \tag{2}$$

- Equivalently, $\nabla f(w^*) = \nabla \ell(w^*, z_i) = 0$ which we call the interpolation condition
- We will discuss the impact of overparameterization and interpolation on convergence and generalization of DL models.



- Recall, for SGD we showed
$$\mathbb{E}\|\nabla f(\hat{w})\|^2 \lesssim \frac{1}{T} + \frac{\sigma^2}{\sqrt{T}} \tag{3}$$
where σ^2 was the variance of the SFO.
- Thanks to interpolation, the variance is not constant, rather decreases as SGD makes progress towards the optimal solution (free variance reduction)

Proposition

Consider $f(w) = \frac{1}{n}\sum_{i=1}^{n} \ell(w, z_i)$ and assume each ℓ is L-smooth. Furthermore assume the interpolation condition holds. Then,
$$\mathbb{E}_{i\sim U[1,\ldots,n]}\|\nabla \ell(w, z_i)\|^2 \leq 2L(f(w) - f^*). \tag{4}$$
That is, the variance of stochastic gradient is bounded by the current suboptimality gap.

- Using this result, one could easily show $\mathbb{E}\|\nabla f(\hat{w})\|^2 \lesssim \frac{1}{T}$.

Proof of Proposition



- Let us calculate the second moment

$$\mathbb{E}_{i\sim U[1,\dots,n]}\|\nabla \ell(w,z_i)\|^2 = \frac{1}{n}\sum_{i=1}^{n}\|\nabla \ell(w,z_i)\|^2 \tag{5}$$

- A consequence of L-smoothness is that if $w_i^* = \arg\min_w \ell(w,z_i)$

$$\|\nabla \ell(w,z_i)\|^2 \leq 2L(\ell(w,z_i) - \ell(w_i^*,z_i)) \tag{6}$$

- Thus over all

$$\mathbb{E}_{i\sim U[1,\dots,n]}\|\nabla \ell(w,z_i)\|^2 \leq 2L\frac{1}{n}\sum_{i=1}^{n}(\ell(w,z_i) - \ell(w_i^*,z_i)) \tag{7}$$

- Using interpolation condition we have $w_i^* = w^*$ for all i. thus, proof is complete by noting the definition of f.

Consequence of Smoothness



- Recall smoothness means

$$f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2, \qquad \forall x, y \in \mathbb{R}^d \tag{8}$$

- Notably, the condition holds for

$$y_x^* = \operatorname*{argmin}_y f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2 \tag{9}$$

- A strongly convex problem. Taking the gradient and setting it to zero

$$y_x^* = x - \frac{1}{L}\nabla f(x) \tag{10}$$

- Substituting in smoothness definition and noting $f(y_x^*) \geq f^* = \min_x f(x)$ establishes

$$\|\nabla f(x)\|^2 \leq 2L(f(x) - f^*) \tag{11}$$

Faster Convergence to Better Solutions



- We showed thanks to interpolation (a property of overparameterized models), SGD-based method enjoy free variance reduction and converge faster to stationary solutions
- Next, we explore a condition that enables faster convergence to global minima.
- The idea is that the loss landscape along the trajectory of SGD-based methods is typically very nice such that these methods only converge to global solutions.
- Notably, along the trajectory, in addition to smoothness, gradient domination or the Polyak-Lojasiewicz (PL) Condition holds:

$$\|\nabla f(w)\|^2 \geq 2\mu(f(w) - f^*), \qquad \mu > 0 \tag{12}$$

- States all stationary solutions are also global solutions (despite nonconvexity).
- Along with the consequence of smoothness, i.e. $\|\nabla f(w)\|^2 \leq 2L(f(w) - f^*)$ leads to faster convergence to global minima

PL condition for Overparameterized Models



- Assume a learning task such that with n data points $z_i = (x_i, y_i)$ where $y_i \in \mathbb{R}$ and $x_i \in \mathbb{R}^p$.
- Consider a parametric DL model $h(w, x)$. Thus, a simple learning problem is

$$\min_w f(w) \quad \equiv \quad \min_w \frac{1}{2n} \sum_{i=1}^{n} \Big(y_i - h(w, x_i) \Big)^2 \quad \equiv \quad \min_w \frac{1}{2n} \|\mathbf{y} - \mathbf{h}(w, \mathbf{X})\|^2 \tag{13}$$

by defining

$$\mathbf{y} = [y_1, \ldots, y_n]^\top \in \mathbb{R}^n, \qquad \mathbf{h}(w, \mathbf{X}) = [h(w, x_1), \ldots, h(w, x_n)]^\top \in \mathbb{R}^n \tag{14}$$

$$\mathbf{X} = [x_1, \ldots, x_n]^\top \in \mathbb{R}^{n \times p} \tag{15}$$

and noting the definition of Euclidean norm

PL condition for Overparameterized Models (Cont'd)



- Let us calculate the gradient of the loss

$$\nabla f(w) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \nabla h(w, x_i))\nabla h(w, x_i) = \frac{1}{n}\nabla \mathbf{h}(w, \mathbf{X}) \cdot (\mathbf{y} - \mathbf{h}(w, \mathbf{X})) \tag{16}$$

  where $\nabla \mathbf{h} \in \mathbb{R}^{d \times n}$ is the gradient of $\mathbf{h}$ i.e., the transpose of its Jacobian.

- Let us calculate the norm square next:

$$\|\nabla f(w)\|^2 = \langle \nabla f(w), \nabla f(w) \rangle = \frac{1}{n}(\mathbf{y} - \mathbf{h}(w, \mathbf{X}))^\top \frac{\nabla \mathbf{h}(w, \mathbf{X})^\top \nabla \mathbf{h}(w, \mathbf{X})}{n}(\mathbf{y} - \mathbf{h}(w, \mathbf{X})) \tag{17}$$

- The matrix $K(w, \mathbf{X}) := \frac{\nabla \mathbf{h}(w, \mathbf{X})^\top \nabla \mathbf{h}(w, \mathbf{X})}{n} \in \mathbb{R}^{n \times n}$ is called the tangent kernel and it is a PSD matrix. Called a kernel since ij entry depends on x_i and x_j

PL condition for Overparameterized Models (Cont'd)



- Let us assume

$$\lambda_{\min}\Big(K(w,\mathbf{X})\Big) = \lambda_{\min}\Big(\frac{\nabla \mathbf{h}(w,\mathbf{X})^\top \nabla \mathbf{h}(w,\mathbf{X})}{n}\Big) = \mu > 0 \tag{18}$$

- Note a necessary condition for this is $d \geq n$, i.e., overparameterization of our DL model.
- Then,

$$\|\nabla f(w)\|^2 \geq \frac{\mu}{n}(\mathbf{y}-\mathbf{h}(w,\mathbf{X}))^\top(\mathbf{y}-\mathbf{h}(w,\mathbf{X})) = 2\mu\cdot\frac{1}{2n}\|\mathbf{y}-\mathbf{h}(w,\mathbf{X})\|^2 = 2\mu f(w) \geq 2\mu(f(w)-f^*) \tag{19}$$

  by definition of f and the fact that $f(w) \geq 0$ for all w.
- This implies if tangent kernel is well conditioned along the trajectory, PL holds and SGD-based method converge very fast to global minima.
- Random initialization can ensure with high probability $\lambda_{\min}\Big(K(w_1,\mathbf{X})\Big) > 0$.
- Thus if the optimization happens near the initialization (e.g., small η and/or L), PL holds for all w_t's with high probability.

Overparameterization and Generalization



- So far we showed the benefits of overparameterization and interpolation in terms of accelerating the training convergence.
- However, overparameterization can further aid with generalization: performing well on test/unseen data
- The main idea is that the training problem of overparameterized model enjoys implicit regularization when solved by SGD-based methods
- Highlighting the crucial role of the methods we talked about, not just in terms of training but also test performance.
- We will explore this idea for linear models $h(w, x_i) = w^\top x_i$ where $d = p > n$. Our task is:

$$\min_w f(w) \quad \equiv \quad \min_w \frac{1}{2n} \sum_{i=1}^{n} \left(y_i - w^\top x_i \right)^2 \quad \equiv \quad \min_w \frac{1}{2n} \|\mathbf{y} - \mathbf{X}w\|^2 \tag{20}$$

- In this case, $K(w, \mathbf{X}) = \mathbf{X}\mathbf{X}^\top$ is independent of model parameters.

Implicit Regularization in Linear Models



- As the system is underdetermined (due to overparameterization), we can perfectly fit all data points such that $f^* = 0$
- Assuming the tangent kernel is positive definite, PL holds for w. Furthermore, the loss is smooth, implying using (S)GD we can find an optimal solution:

$$\lim_{t\to\infty} f(w_t) = 0, \qquad w_{t+1} = w_t - \eta\nabla f(w_t), \quad \eta \leq 1/L \tag{21}$$

- Thus, if $\hat{w} = \lim_{t\to\infty} w_t$, we have that $\mathbf{y} = \mathbf{X}\hat{w}$ exactly
- We will then show among all possible solutions w^*, GD's solution $\hat{w}$ is the one closest to the initialization w_1, i.e.,

$$\|w_1 - \hat{w}\|^2 \leq \|w^* - w_1\|^2, \qquad \forall w^* \in \arg\min_w f(w), \qquad w^* \neq \hat{w} \tag{22}$$

- So, implicitly, GD is solving an ℓ_2 regularized problem:

$$\min_w \|w - w_1\|^2 \quad s.t. \quad \mathbf{y} = \mathbf{X}w \qquad \equiv \qquad \min_w f(w) + \frac{\lambda}{2}\|w - w_1\|^2 \tag{23}$$

Some Linear Algebra Overview



- Range: The range of a matrix $\mathbf{X}$ is the set of all vector a that can be written as a linear combination of columns of $\mathbf{X}$. Notably, any vector $a = \mathbf{X}u$ is in the range of $\mathbf{X}$.
- Null-space: the null-space of a matrix $\mathbf{X}$ is the set of all vectors b such that $\mathbf{X}b = 0$.
- For a matrix $\mathbf{X}$ we have that Range($\mathbf{X}^\top$) and Null($\mathbf{X}$) are orthogonal:

$$b \in \text{Null}(\mathbf{X}) \quad \Leftrightarrow \quad \text{Proj}_{\text{Range}(\mathbf{X}^\top)}(b) = 0, \qquad a \in \text{Range}(\mathbf{X}^\top) \quad \Leftrightarrow \quad \text{Proj}_{\text{Null}(\mathbf{X})}(a) = 0$$

- For our problem where $\mathbf{y} = \mathbf{X}w$, as $d > n$, $\mathbf{X}$ is a fat matrix and has a non-trivial null-space.
- The vector $\bar{w} = \mathbf{X}^\dagger \mathbf{y}$ is in the range of $\mathbf{X}^\top$. $\mathbf{X}^\dagger = \mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}$ is the pseudo-inverse of $\mathbf{X}$
- The gradient $\nabla f(w) = \frac{1}{n}\mathbf{X}^\top(\mathbf{X}w - \mathbf{y})$ is in the range of $\mathbf{X}^\top$.
- The set of all solutions is

$$S = \{w^* | w^* = \bar{w} + b,\ b \in \text{Null}(\mathbf{X})\} \tag{24}$$

and notably $\text{Proj}_{\text{Range}(\mathbf{X}^\top)}(w^*) = \bar{w}$ for all w^*

Implicit Regularization in Linear Models (Cont'd)



- Given the update $w_{t+1} = w_t - \eta \nabla f(w_t)$ and the fact that $\nabla f(w_t) \in \text{Range}(\mathbf{X}^\top)$ we can show

$$\text{Proj}_{\text{Null}(\mathbf{X})}(\hat{w}) = \text{Proj}_{\text{Null}(\mathbf{X})}(w_t) = \text{Proj}_{\text{Null}(\mathbf{X})}(w_1), \qquad \hat{w} = \lim_{t \to \infty} w_t \tag{25}$$

- Since $\text{Range}(\mathbf{X}^\top)$ and $\text{Null}(\mathbf{X})$ are orthogonal, for any $w^* = \bar{w} + b \neq \hat{w}$

$$\begin{aligned}
&\|w^* - w_1\|^2 = \|\text{Proj}_{\text{Null}(\mathbf{X})}(w^*) - \text{Proj}_{\text{Null}(\mathbf{X})}(w_1)\|^2 + \|\text{Proj}_{\text{Range}(\mathbf{X}^\top)}(w^*) - \text{Proj}_{\text{Range}(\mathbf{X}^\top)}(w_1)\|^2 \\
&\geq \|\text{Proj}_{\text{Range}(\mathbf{X}^\top)}(w^*) - \text{Proj}_{\text{Range}(\mathbf{X}^\top)}(w_1)\|^2 \\
&= \|\text{Proj}_{\text{Range}(\mathbf{X}^\top)}(\hat{w}) - \text{Proj}_{\text{Range}(\mathbf{X}^\top)}(w_1)\|^2 \\
&\overset{(b)}{=} \|\text{Proj}_{\text{Range}(\mathbf{X}^\top)}(\hat{w}) - \text{Proj}_{\text{Range}(\mathbf{X}^\top)}(w_1)\|^2 + \|\text{Proj}_{\text{Null}(\mathbf{X})}(\hat{w}) - \text{Proj}_{\text{Null}(\mathbf{X})}(w_1)\|^2 \\
&= \|\hat{w} - w_1\|^2
\end{aligned} \tag{26}$$

- Orange term in (b) is zero due to eq. (21)



- Consider again the general task

$$\min_w f(w) \quad \equiv \quad \min_w \frac{1}{2n} \sum_{i=1}^{n} \Big(y_i - h(w, x_i) \Big)^2 \quad \equiv \quad \min_w \frac{1}{2n} \|\mathbf{y} - \mathbf{h}(w, \mathbf{X})\|^2 \tag{27}$$

- To apply our previous results, one typically linearizes the model around the initialization w_1 to get a linear model:

$$\mathbf{h}^l(w, \mathbf{X}) = \mathbf{h}(w_1, \mathbf{X}) + \nabla \mathbf{h}(w_1, \mathbf{X})^\top (w - w_1) \tag{28}$$

- This leads to a linear regression task

$$\min_w \frac{1}{n} \|\tilde{\mathbf{y}} - \nabla \mathbf{h}(w_1, \mathbf{X})^\top (w - w_1)\|^2, \qquad \tilde{\mathbf{y}} = \mathbf{y} - \mathbf{h}(w_1, \mathbf{X}) \in \mathbb{R}^n \tag{29}$$

- Our results then hold for the linearized model. And thanks to overparameterization and using SGD-based methods with small learning rates, the approximation error is small (e.g. $\mathcal{O}(n/\sqrt{d})$).